

Unit Topic: Energy

Essential Question: How does energy relate to forces?

Concept 1: <i>Nature of Energy</i>	Concept 2: <i>Conservation of Energy</i>	Concept 3: <i>Thermal Energy</i>	Concept 4: <i>Work and Power</i>
Objectives: <i>Discern if an object has kinetic or potential energy, based on a description.</i> Vocabulary: Energy Kinetic energy Potential energy Gravitational potential energy Practice: 1. Explain the differences between the 3 types of potential energy. 2. What is the energy of a 4 kg apple that is sitting on a 2 m high tree branch? 3. What is the energy of a baby who weighs 20 N sitting in a high chair 1.5 m high? 4. What is the kinetic energy of a bike with a mass of 16 kg traveling at 4 m/s? 5. How high is a 0.5 kg squirrel sitting if it has 36 J of energy?	Objectives: <i>Given a picture or a description, explain the energy conversion occurring in an object.</i> Vocabulary: Law of Conservation of Energy Electromagnetic energy Mechanical energy Practice: 6. Give an example of where you would find each of the following types of energy: radiant, thermal, electrical, sound, nuclear, and electromagnetic. 7. What is the relationship between kinetic and potential energy of a falling object? 8. Determine the energy conversion for the following: a. Light bulb b. Firework c. Flute d. Leaf 9. Draw a picture of a pendulum or someone hitting a golf ball, showing how GPE, KE, and ME change throughout the object's motion. Make your labels very clear.	Objectives: <i>Identify materials as conductors or insulators.</i> <i>Given a picture or a scenario, explain how thermal energy is being transferred.</i> Vocabulary: Thermal energy Temperature Heat Insulator Conductor Specific heat Practice: 10. Explain the relationship between temperature and kinetic energy. 11. Explain why there is no such thing as "cold". 12. Describe how thermal energy flows when a person is sitting in a chair at their desk. 13. Explain the difference between the 3 types of thermal energy transfer. Include an example of each. 14. List examples of materials that would classify as conductors vs. insulators. 15. How much heat is needed to warm 0.072 kg of gold from 20°C to 90°C if the specific heat of gold is 136 J/kg/°C?	Objectives: <i>Use Newton's 2nd Law to complete work and power calculations.</i> Vocabulary: Work Power Practice: 16. A force of 85 N is used to push a box along the floor a distance of 15 m. How much work was done? 17. 150 J of work was done to lift a crate with a force of 20 N. How far was the crate lifted? 18. How much power is used if a force of 90 N is used to push a box a distance of 30 m in 10 s? 19. A go-cart and rider have a mass of 100 kg. If the cart accelerates at 5 m/s² during a 25 m sprint, how much work did the cart do? 20. A 10 kg rock falls from a height of 18 m and lands on a guy about 3 s later. How much power did it hit him with?

Concept 1 Notes: Nature of Energy

Questions/Vocabulary

Energy

Kinetic Energy

Example #1: *A jogger with a mass of 60 kg is moving at a speed of 3.0 m/s. What is the jogger's kinetic energy?*

270 J

Practice Time!

1. 120 J

2. 5.42 kg

3. 50 m/s

Potential Energy

Elastic potential energy

Chemical potential energy

Gravitational potential energy

Example #2: *How much gravitational potential energy does a 5 kg rock have if it is sitting on the edge of a cliff that is 10 m high?*

490 J

Practice Time!

1. 7,056,000 J

2. 0.914 m

3. 216 J

Summary Concept 1

Explain the difference between KE and PE, including the variables that affect each.

Practice: Kinetic and Potential Energy #1

Answer the following questions. Make sure to show all work to receive credit. You may need a separate sheet of paper.

1. Lauryn serves a volleyball with a mass of 2.1 kg. The ball leaves her hand with a speed of 30 m/s. Find the energy of the ball.

945 J

2. A cinder block sits on a platform 20 m high. If it has a mass of 8 kg, find its energy.

1,568 J

3. A baby's car seat sits on top of a bench 2 m tall. The baby and the car seat have a weight of 12 N. Find the energy of the baby and its car seat.

24 J

4. A 1,120 kg car is travelling with a speed of 40 m/s. Find its energy.

896,000 J

5. A bell hanging at the top of a tower has 8,550 J of energy. If it has a mass of 20 kg, find the height of the bell.

43.6 m

6. **Challenge:** What is the mass of an object that has 324 J of energy and is moving at a speed of 9 m/s?

8 kg

Practice: Kinetic and Potential Energy #2

Answer the following questions. Make sure to show all work to receive credit. You may need a separate sheet of paper.

1. What is the potential energy of a 2 kg plant that is on a windowsill 1.3 m high?

25.5 J

2. What is the kinetic energy of a 7.26 kg bowling ball that is rolling at a speed of 2 m/s?

14.52 J

3. An apple has 6 J of energy. The apple sits on a counter 3 m high. What is the mass of the apple?

0.204 kg

4. Two objects are being lifted by a machine. One object has a mass of 2 kg, and is lifted at a speed of 2 m/s. The other has a mass of 4 kg and is lifted at a rate of 3 m/s.

- a. Which object has more kinetic energy while it is being lifted?

The 2nd object (4 kg object)

- b. What does this tell you about the variable(s) that impact kinetic energy?

That mass and velocity impact amount of kinetic energy

- c. Which object had more potential energy when it was lifted a distance of 10 m?

The 2nd object (4 kg object)

- d. What does this tell you about the variable(s) that impact gravitational potential energy?

That mass, acceleration due to gravity, and height impact amount of GPE

5. You are sitting in a cart at the top of a hill on a roller coaster. You have 10,500 J of GPE. If your current weight is 539 N, how tall is the hill you are sitting on?

19.5 m

6. What is the potential energy of your 3 kg puppy that is sitting in the grass in your backyard?

0 J

7. In a lab investigation, one group of students (group A) measures the speed of a 0.1 kg toy car to be 2.5 m/s at the bottom of a ramp. Another group of students (group B) measures the speed of the car to be 3 m/s at the bottom of the ramp. The car's starting position at the top of the ramp is 1 m high.

- a. What is the potential energy of the car at the beginning of the experiment before its speed is measured?

0.98 J

- b. Calculate the kinetic energy of the car for group A.

0.3125 J

- c. Calculate the kinetic energy of the car for group B.

0.45 J

8. **Challenge:** You are at the Grand Canyon, standing at the edge of a ledge 1,857 m high. You have a mass of 61 kg. You decide to take a selfie to share with your science teacher when you get home which causes you to wonder...

- a. How much gravitational potential energy do you have standing at the edge of this cliff?

1,110,114.6 J

- b. If you were to trip and fall mid-selfie, your stored potential energy would be converted to kinetic energy, as you are in motion. Assume that all of the GPE you calculated in part A would be converted to kinetic energy when you fall. If so, how fast would you be falling?

190.8 m/s

Lab Activity: Exploring the Motion of a Pendulum

Guiding Question: How does the change in motion of a pendulum ride at a fair or amusement park affect energy?

Purpose: To determine the relationship between energy and motion by investigating an amusement park ride.

Introduction: Have you ever been to a fair or an amusement park and seen a Viking ship ride like the one pictured to the right? Better yet, have you ever been on one of these rides? If you haven't, these rides basically act like giant pendulums, swinging you back and forth. If you have, you can attest that as the ride cranks you up to the side, there is a moment of stillness (and terror) right before it drops you in a giant swinging motion, back and forth until the ride comes to an end. Rides like this can tell us a lot about the relationship between motion and energy, which is why we will be exploring them today.

Hypothesis: *Write a prediction (it can be in an informal format for this activity) about the relationship between energy and motion that you think exists on rides such as this.*



Answers will vary

Materials: (per group)

- 3 washers
- 1 piece of yarn
- 1-2 meter sticks
- 1 stopwatch or timer
- Digital scale

Procedure:

1. Find the mass of 1 washer using the digital scale. Record in Data Table 1.
2. Find the mass of all 3 washers together using the digital scale. Record in Data Table 2.
3. Measure and cut your piece of yarn to be about 0.6 m (60 cm) long.
4. Tie one end to 1 washer. Lay flat. Now measure the length from the bottom of the washer to the other end of the yarn. Trim off the end until the entire length of the yarn AND washer is 0.5 m (50 cm). You now have your first pendulum! This will be used to simulate the motion of rides at the amusement park, like the one pictured above.
5. Group member #1 should be standing, holding the meter stick vertically, with the zero end resting on the ground. He/she should also be holding the string end of the pendulum at 0.5 m high. If being held correctly, the washer should be *almost* touching the ground, but not quite (we don't want it to hit the ground and have that impact its motion, but we do want it to swing as close to the ground as possible.)
6. Group member #2 should be sitting on the ground with the stopwatch, ready to take the time.
7. Group member #3 should pick up the washer, being careful not to pull it out of person #1's hand. It should be lifted outward so that it is parallel to the ground and perpendicular to the meter stick. Measure the height from the washer to the ground with another meter stick (if you have one.) If it is being held correctly, you should get a height of 0.5 m.
8. On "go", person #3 should let go of the washer and person #2 should start the stopwatch, observing the motion of the pendulum. Person #2 should stop the stopwatch when the pendulum passes just above the ground before continuing in its path upwards. It may take a few tries to get it. Record the time in Data Table 1. Repeat for 3 trials.
9. Repeat steps 3-8, this time tying all 3 washers to your pendulum. Record results for all 3 trials in Data Table 2.
10. Calculate average times in Data Tables 1 and 2. Then answer all analysis questions.

Results Data will vary for each group

Data Table 1: Pendulum with 1 Washer

Mass =	
Trial #	Time it took to reach bottom of path (s)
1	
2	
3	
Average	

Data Table 2: Pendulum with 3 Washers

Mass =	
Trial #	Time it took to reach bottom of path (s)
1	
2	
3	
Average	

Note: Remember this is an inquiry investigation - they haven't learned about energy conversions yet! So be gracious with your grading and know you may need to help them through this, but I think it is good for them to struggle some as they reason through the analysis questions!

Analysis

1. What type of energy did the pendulum have before person #2 let go? Calculate this for both pendulums.

GPE

Answers will vary based on data

2. Referring to your calculations from #1, which pendulum had more initial energy? Explain why this makes sense.

Should be the 3 washer pendulum because it should have a greater mass

3. How could we have made both pendulums have more initial energy?

Either added more mass or dropped them from a greater height

4. What type of energy did the pendulum have after person #2 let go? Explain how you know.

Kinetic energy, because the pendulum is now in motion and all objects in motion have kinetic energy

5. When in the pendulum's swing should it have had the most gravitational potential energy? Explain how you know. You may draw a picture if that helps.

When it had the greatest height, which is at the beginning

6. When in the pendulum's swing should it have had the most kinetic energy? Explain how you know. You may draw a picture if that helps.

It should have the most kinetic energy when it is moving the fastest (or having the lowest GPE) which should be at the bottom of its swing before it starts swinging upwards again

7. Calculate the peak kinetic energy for both pendulums in their swings.

Answers will vary, but should be able to find the peak velocity at the bottom of the swing using acceleration and average time to rearrange for final velocity (initial would be zero). (You may need to walk CP students through this part as a class, because these are some serious inquiry-based questions!)

8. Referring to your calculations from #7, which pendulum had more kinetic energy? Explain why this makes sense.

The pendulum with the 3 washers should have the most kinetic energy because #1 it has more mass and #2 it started with more GPE

9. How could we have made both pendulums have more kinetic energy?

#1 added more mass or #2 dropped them from a greater height so they would start off with more GPE (that will convert to KE as it falls, but they won't understand this yet!)

10. In Concept 2 we will learn about the Law of Conservation of Energy, which states that energy can never be created or destroyed, it can only change forms. Where do you think the energy you described in #4 came from?

All of the kinetic energy came from the initial gravitational potential energy

****Thinking Ahead:** What do you think the relationship is between kinetic and potential energy in systems like the pendulum ride we investigated today?

As one goes up, the other goes down! (Ex. as the pendulum fell, GPE decreased and KE increased)

Concept 2 Notes: Conservation of Energy

Questions/Vocabulary

Law of Conservation of Energy

Forms of Energy

Radiant

Thermal

Electrical

Sound

Nuclear

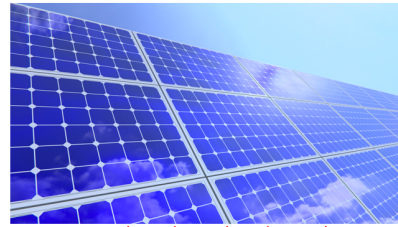
Electromagnetic

Energy Transfers

What are some examples of energy transfers/conversions?



chemical to light/thermal/sound



solar to electrical



electrical to light



electrical to electromagnetic
(and light and sound)



solar to chemical



kinetic to sound

Mechanical energy



Summary Concept 2

Connect what we explored in the pendulum lab activity to what we learned in this concept.

Lab Stations: Energy Conversions

Travel to each station in the room and observe the pictures or video clips. Infer the energy transformation that is occurring. You do not have to go in order, just make sure you get to every station!

<u>Station 1</u> kinetic to electrical	<u>Station 2</u> chemical to radiant/thermal
<u>Station 3</u> sound to electromagnetic to sound	<u>Station 4</u> electrical to thermal (and kinetic)
<u>Station 5</u> chemical to thermal/kinetic	<u>Station 6</u> kinetic to sound
<u>Station 7</u> chemical (battery) to electromagnetic	<u>Station 8</u> electrical to radiant/sound/electromagnetic
<u>Station 9</u> elastic potential to kinetic	<u>Station 10</u> electrical to thermal (and kinetic!)
<u>Station 11</u> kinetic to chemical to radiant/thermal	<u>Station 12</u> electromagnetic to electrical

Lab Investigation: Law of Conservation of Energy

Guiding Question: How does energy transform, with respect to the law of conservation of energy, when a ball is in motion?

Purpose: To investigate energy transformations in a falling ball with respect to the law of conservation of energy.

Hypothesis:

Make a prediction about which ball you believe will have the most initial energy and will retain the most energy when it bounces.

answers will vary

Materials:

- 4 different balls
- Meter stick
- Digital scale

Procedures:

1. Find the mass of one of the balls. Record in Data Table 1.
2. Have one member of the group hold the meter stick upright, with the zero end on the floor. Hold the ball at the top of the meter stick (1 m from the ground.)
3. Another member of the group should squat down to eye level with the meter stick. When the ball is dropped, measure the height of the ball after the 1st bounce. Record in Data Table 2.
4. Drop the ball again, this time recording the height of the 2nd bounce. Make sure the same person measures the height each time for consistency. Re-do the drop if you don't think an accurate measurement was made.
5. Drop the ball again and record the height of the 3rd bounce.
6. Repeat steps 1-5 for all four balls.

Results

Data Table 1: Masses of different balls

Type of Ball	Mass (g)

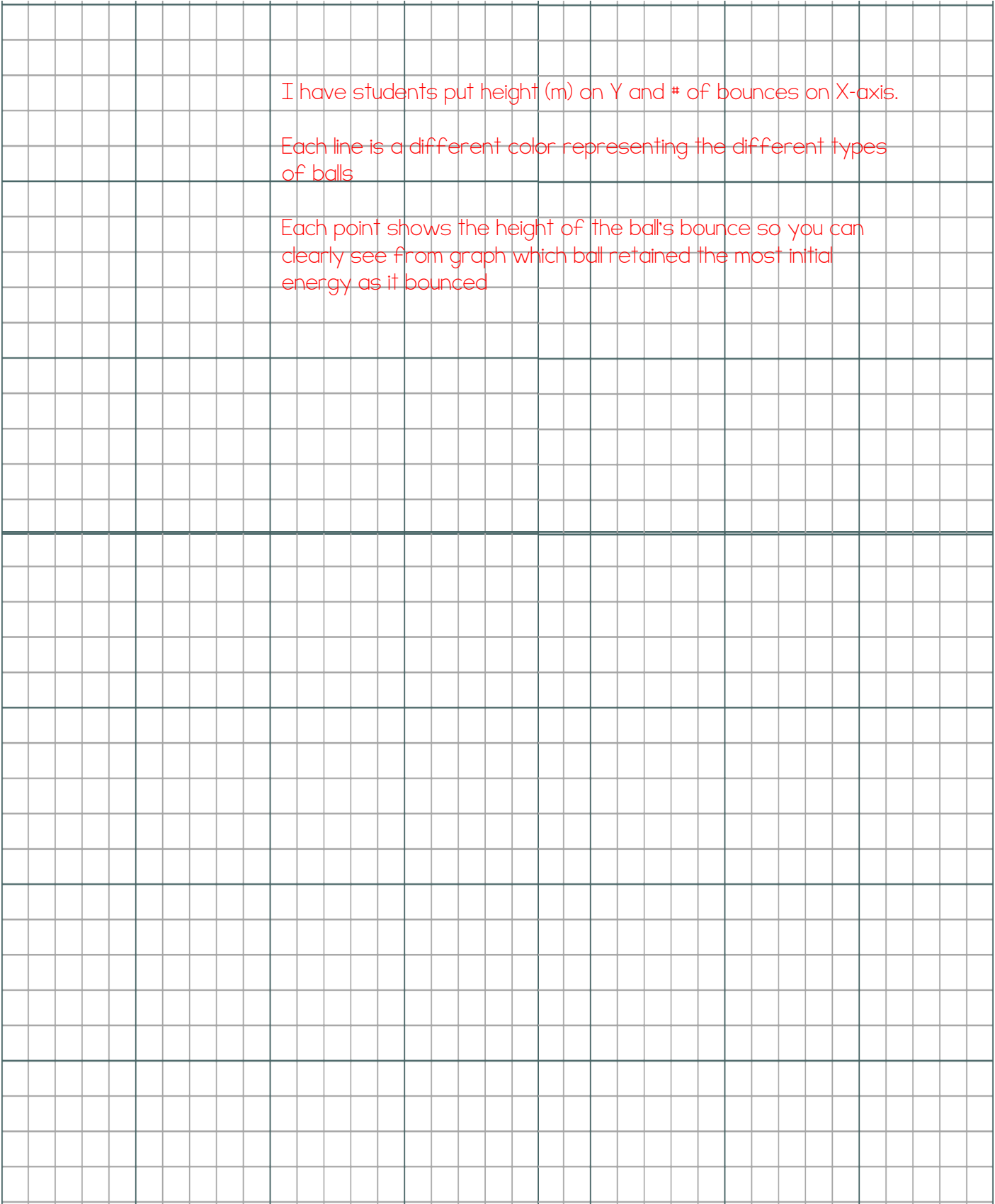
Data will vary depending on the types of balls you provide for your students to use

Data Table 2: Height of bounces

Type of Ball	Height of 1 st Bounce (m)	Height of 2 nd Bounce (m)	Height of 3 rd Bounce (m)

Make sure they measure in m and not cm!! (good time to stop and review metric units and conversions!)

Make a line graph below showing the height of the different bounces for each ball. You will need to make sure to include a title, labeled X and Y-axis with units (if needed), and a key that shows which colors correspond to which balls.



Analysis: Answer the following analysis questions below.

1. What type of energy did the balls have just before they were released? Calculate this energy for **each** of the four balls below.

GPE

Calculations will vary based on balls used

2. Consider the energy transformation as each ball fell. What form did the energy from #1 convert to? What energy did it convert to when it hit the ground? Explain the energy transformation of a bouncing ball below.

Kinetic when it is dropped

Elastic potential when it hits the ground

Then kinetic again as it moves back upwards

3. Refer to your graph and explain what it shows you.

Answers will vary

Should mention something about the height of the different balls compared to each other and how the height of each bounce always decreased

4. If all of the initial energy was converted as the ball fell and hit the floor, what should we have seen when it bounced back? What did you actually see?

It should have bounced back to the original height of 1 m, according to the law of conservation of energy. Actually saw that each bounce was a lower height, suggesting that not all energy was conserved.

5. Why can a dropped ball never bounce higher than the height from which it was dropped?

Because according to the law of conservation of energy, energy can not be **gained** (or destroyed)

6. Refer to your answers to #4 and 5. What does this tell you about the Law of Conservation of Energy? Can it be violated? Explain your reasoning.

It doesn't mean that the law can be violated or isn't true. It just means that the energy must be being converted to other forms besides kinetic energy!

7. Refer to your hypothesis. Which ball did you expect to have the most initial energy? Which actually did? Which did you expect to retain the most energy? Which ball actually did? Explain your reasoning, with reference to your data.

Answers will vary depending on hypotheses.

Note: Students may have predicted that the ball with most mass would have the greatest amount of GPE and thus the greatest height of bounce back BUT this may not be seen due to some balls have greater elasticity thus having a more efficient conversion from kinetic to elastic potential energy and back to kinetic

8. Explain what other forms of energy the initial energy may have converted to.

Sound (you can hear it bounce!) and **thermal** (due to friction)

****Note**: This is a hard question for them to process through**

9. Error analysis: Write out three ways your data may not be completely accurate. Think about ways that your numbers collected from each trial may not be consistent with each other. Include a specific solution for each error to prevent these problems in the future for other students who do this lab.

Answers will vary

10. Application: How would we may have seen different results in a different situation, such as using different types of balls than the ones provided or dropping the balls on carpeted floor vs. tile? Explain based on what you know about energy.

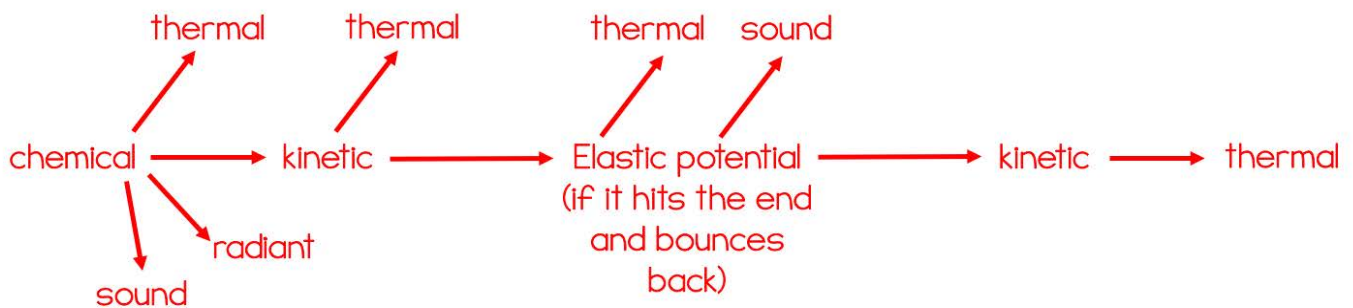
Answers will vary

But for example, carpeted floor should not bounce as well because more energy gets converted to thermal due to increased friction

Activity: Alcohol Rocket Demonstration

Your teacher is going to create a bottle rocket powered by ethanol. When ignited, energy will transform and multiple energy conversions will occur during the rocket's motion. Observe the motion of the rocket. Then discuss the energy conversions witnessed with your lab partner. Below, draw and label a diagram that represents the transfers of energy that occur from start to finish of the rocket's launch.

Let them get creative in how they draw this! But it could be something like:



Concept 3 Notes: Thermal Energy

Questions/Vocabulary

Thermal Energy

Temperature

Heat

Transfer of Thermal Energy

Conduction

Convection

Radiation

Materials

Conductors

Insulators

Specific Heat

Example #1: *A wooden block has a mass of 20 kg and a specific heat of 1,700 J/kg °C. Find the change in thermal energy of the block as it warms from 15°C to 25°C.*

340,000 J

Practice Time!

1. 306,000 J

2. 0.15 kg

Summary Concept 3

List some of the ways you witness thermal energy transfer every day in your life.

Practice: Thermal Energy Calculations

Answer the following questions. Make sure to show all work for the math problems to receive credit. You may need a separate sheet of paper.

1. Explain the relationship between temperature, energy, and motion of particles in an object.

Temperature is the average kinetic energy of particles in an object

So, the higher the temperature, the greater the kinetic energy of the particles, therefore the faster the particles are moving

2. Referencing the reasoning you used from #1, explain the difference between objects that feel hot and those that feel "cold".

Objects that feel hot have more heat, because they have a greater temperature and thus a greater kinetic energy in their particles. Objects that feel "cold" have less heat, because they have a lower temp. and thus lower KE in their particles.

3. You've been waiting for the bus and your hands become cold. When you get onto the bus and sit down, you put your hands under your legs to warm up. After a while your hands feel warmer but your legs feel colder. Explain this with regards to what you know about thermal energy transfer.

Your body is warmer than your hands, thus has a higher temperature and more heat. Heat flows from higher temperature objects to lower temperature objects, so heat will leave your legs and move to your hands, thus you feel the absence of heat in your legs.

4. How much energy must be absorbed by water with a mass of 0.5 kg in order to raise the temperature from 30°C to 65°C? Note: Water has a specific heat of 4,190 J/kg °C.

73,325 J

5. How much heat is needed to warm .052 kg of gold from 30°C to 120°C? Note: Gold has a specific heat of 136 J/kg °C.

636.5 J

6. A 9.5 kg outdoor copper sculpture heats up during the day from 24°C to 78°C. How much energy was absorbed? Note: Copper has a specific heat of 390 J/kg °C.

200,070 J

7. **Challenge:** If it takes 820 Joules of heat to warm a sample of zinc from 0°C to 50°C, what would be the mass of the zinc? Note: Zinc has a specific heat of 380 J/kg °C.

0.043 kg

Note: This is a full inquiry investigation so let them explore and struggle! There is no one right way to do this so this "answer key" is very limited.

Inquiry Investigation: Insulation Materials

Introduction: Rebecca loves to have ice cold water on her bedside table to drink when she wakes up in the middle of the night. She has been learning in physical science class about thermal energy transfer, and wants to figure out a way to create a homemade insulator for her ice water. Can you help her?

Guiding Question: *Write an overarching question to try to answer for this lab.*

Answers will vary.

Ex. What materials make the best insulation for ice water?

Purpose: *Re-write your question as a "To determine..." statement.*

Answers will vary.

Ex. To determine what materials make the best insulation for ice water.

Background Knowledge: *What do you know about thermal energy transfer? What makes her ice cube melt? How can that be prevented? What materials make good insulators? What materials do not make good insulators?*

Students should know...

Thermal energy (heat) transfers from high temp to low temp

Her ice keeps melting because her cup is allowing heat into the cup

It can be prevented by creating an insulating cup that keeps heat out

Good insulators = plastic, foam, rubber, etc.

Bad insulators (aka good conductors) = metal, air, skin

Make a Plan:

a. *How could you investigate this?*

b. *What materials do you want to test out?*

c. *What else will you need to investigate this guiding question?*

d. *What will be your independent variable in this investigation?*

Answers will vary.
Ex. different materials

e. *What will be your dependent variable?*

Answers will vary.
Ex. Temperature of water (which will reflect how much heat was let in)

f. *What will be your control group?*

Ex. A normal cup, maybe just a clear plastic one

g. *What will be your experimental group(s)?*

Ex. All of the different cups they build with insulating materials

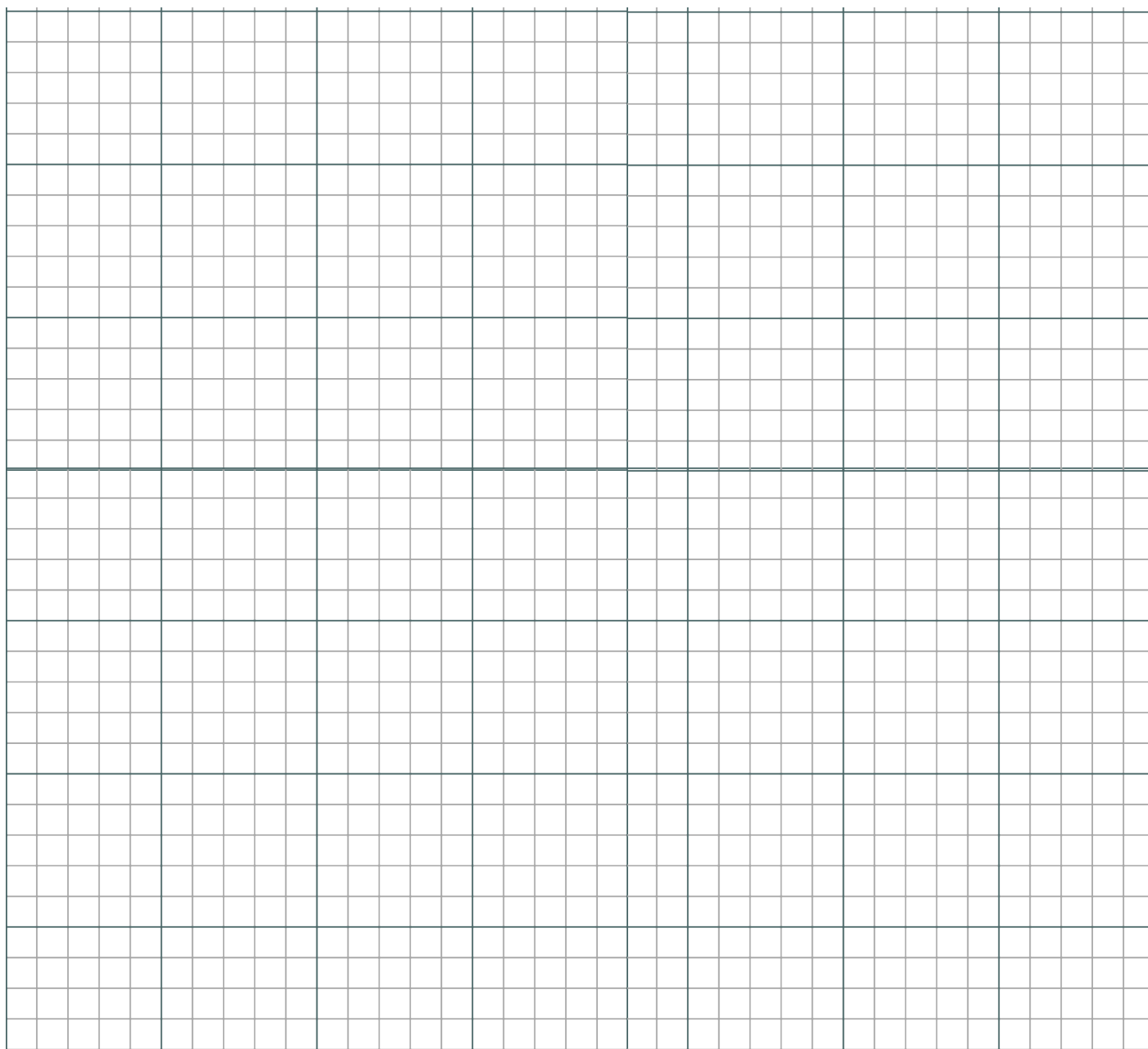
h. *What constants do you need to plan for in your procedures?*

Hypothesis: *Based on the background knowledge you wrote out above, make a prediction about the material you believe will make the best insulator.*

Materials: *Now that you have planned your investigation, make a specific **bulleted list** of what you will **need** to conduct this investigation.*

Procedures: *Now that you have planned your investigation, make a specific **numbered list** of what you will need to **do** for this investigation.*

Results: *Set up a data table for how you want to collect your data. After conducting the experiment, create a graph that best represents the data you've collected.*



Analysis: Write a paragraph answering the following questions.

- What does your data show you? Refer to your graph in your answer.
- In what ways could you have improved your experimental design? List three specific weaknesses or errors and three specific solutions for how these could be fixed/prevented in the future.

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

Conclusion: Write a paragraph answering the following questions.

- Does the data cause you to support or reject your hypothesis? Explain how you know.
- What did you learn about thermal energy transfer and insulators from this investigation?
- How would what you learned be helpful knowledge in the real world? (aka life outside of science class!)

[illegible]

Concept 4 Notes: Work and Power

Questions/Vocabulary

Work

Two conditions must be met:

Example #1: *You move a 75 kg refrigerator 35 m. This requires a force of 90 N. How much work, in joules, was done while moving the refrigerator?*

3,150 J

Practice Time!

1. 2 J

2. 19.6 J

3. 25 m

4. 8 N

Power

Example #2: *It took 150 seconds to move a refrigerator. You did 3,150 Joules of work in the process. How much power was required to move the refrigerator?*

21 J/s or 21 W

Summary Concept 4

Why does it make sense that work is measured in Joules? How is power similar to speed?

Practice: Work and Power Calculations

Answer the following questions. Make sure to show all work to receive credit. You may need a separate sheet of paper.

1. It took 500 N of force to push a car 4 meters. How much work was done?

2,000 J

2. A force of 15 N is used to push a box along the floor a distance of 3 meters. Find the amount of work done.

45 J

3. A force of 100 N was needed to lift a rock. A total of 150 J of work was done. How far was the rock lifted?

1.5 m

4. Max exerted a force of 9,000 N on a stalled car but was unable to move it. How much work did he do?

0 J

5. A 5 kg rock is lifted 2 m. Find the amount of work done.

98 J

6. How much power is used if a force of 35 N is used to push a box a distance of 10 meters in 5 seconds?

70 W

7. If 270 watts of power is used in 42 seconds, how much work was done?

11,340 J

8. How long does it take to do 278 J of work with 17 W of power?

16.35 s

9. A pulley is used to lift a piano weighing 1,000 N. The piano is lifted 3 m in 60 s. How much power is used?

50 W

10. **Challenge:** If it takes 726 watts of power to move an object 36 m in 14 s, what is the mass of the object?

28.8 kg

Research and Report: Simple Machines

What do we already know about this topic?

Topic: _____

Record notes from your research to share with your peers.

Record below what you learn from your peers after they share their findings.